

Bank Fraud Detection Report

This report presents a comprehensive Exploratory Data Analysis (EDA) on one million bank transactions, examining behavioral patterns, demographic distributions, and channel-level anomalies to surface key indicators of fraudulent activity. Findings are intended to guide downstream machine-learning model development and operational fraud-prevention strategies.

Dataset overview

The dataset comprises 1,000,000 transaction records described by 26 features spanning payment method, merchant category, device type, customer demographics, and transaction amount. The binary target variable is fraud exhibits a strong class imbalance, a fundamental modelling challenge explored throughout this report.

Attributes	Detail
Rows	1,000,000
Columns	26
Target Variable	Is Fraud(binary)
Fraud Cases	55,255 (5.53%)
Non-Fraud Cases	944,745 (94.47%)

This means the dataset is imbalanced, which is an important finding for your report.

Visualizations & Observations

Transaction Amount Outlier Analysis

A boxplot of transaction amount was used to characterize the spread and detect outliers across the full dataset.

- The distribution is heavily right-skewed, compressing the interquartile box toward the lower end of the scale.
- A large volume of extreme high-value outliers extends well into the tens-of-thousands-of-dollars range.

- Legitimate everyday transactions cluster within a predictable, low-to-moderate price range.

Insight: Ultra-high transaction values are a disproportionately strong signal of potential fraud. Any transaction significantly above the 99th-percentile threshold warrants automated review.

Customer Age Distribution

A histogram with an overlaid Kernel Density Estimate (KDE) was used to profile the age distribution of account holders.

- Age follows a near-normal distribution peaking in the 35–55 year-old cohort.
- Participation tapers sharply among adults under 20 and over 70.

Insight: The core user base is working-age adults. Sudden fraud surges targeting younger or elderly sub-populations represent statistically anomalous patterns that risk models should flag explicit.

Fraud Distribution by Payment Method

A grouped bar chart (crosstab) compared legitimate versus fraudulent transaction counts across all payment method categories.

- Legitimate transactions dominate every category due to the class imbalance, but the relative height of fraud bars reveals meaningful variation.
- Certain payment channels exhibit a substantially higher fraud-to-legitimate ratio than others.

Insight: Fraudsters gravitate toward payment channels offering faster execution or weaker immediate

verification. Channel-specific velocity rules and step-up authentication would reduce exposure.

Fraud by Merchant Category

A count-plot segmented by merchant category and colored by is fraud highlighted sector-level risk concentrations.

- High-resale categories: luxury goods, consumer electronics, gift cards: exhibit disproportionately elevated fraud frequencies.
- Essential services and low-resale retail display stable legitimate volumes with minimal fraud spikes.

Insight: Fraudulent transactions are strategically concentrated in merchant sectors where items can be quickly liquidated or resold. Merchant-category risk scores should be incorporated as a core feature in any fraud classifier.

Fraud by Device Type

A count-plot tracking device type against fraud status examined the relationship between the transaction interface and fraud likelihood.

- Transactions originate from web browsers, native mobile apps, and alternative device endpoints.
- Certain interfaces particularly mobile emulators and unsecured web entry points harbor a significantly higher ratio of fraudulent attempts compared to secure native-app sessions.

Insight: Device type is a robust technical proxy for risk. Device fingerprinting, emulator detection, and session-integrity checks should be prioritized in the fraud-prevention stack.

Key Findings Summary

The table below consolidates the principal findings from the EDA and their associated risk severity.

No	Finding Risk Level	Risk Level
1	Heavy right-skew & extreme outliers in transaction amounts	High
2	Middle-aged (35–55) users form the core demographic	Medium
3	Certain payment channels show disproportionate fraud	High
4	Luxury/electronics merchant categories are high-risk target	High
5	Device type is a strong technical proxy for risk	Medium
6	Severe class imbalance (5.53% fraud	High

Class Imbalance

With only 5.53% of records labelled fraudulent, standard accuracy metrics are misleading. A naïve classifier predicting 'legitimate' for every transaction achieves >94% accuracy while catching zero fraud. Techniques such as SMOTE oversampling, class-weighted loss functions, and precision-recall optimization are mandatory for any downstream model.

Strategic Targeting by Fraudsters

Fraud is not randomly distributed; it is concentrated in high-value transaction segments, specifically merchant categories, and vulnerable payment channels. This deliberate clustering

means ensemble models trained on interaction features (e.g., amount $\times$ merchant category) will materially outperform single-feature rules.

Anomalous Transaction Amounts

Most of the legitimate activity is bounded within a predictable low-to-moderate value range. Extreme outlier transactions, particularly those well beyond the 99th percentile, are a high-signal fraud indicator. Log-transformation of the amount feature and outlier-specific sub-models are recommended.

Conclusion

This Exploratory Data Analysis has established a clear evidence base for the drivers of fraudulent activity within the bank transaction dataset. Three structural realities define the modelling challenge ahead: a severe class imbalance requiring careful sampling strategy, a strategic concentration of fraud in specific merchant categories and payment channels, and the outsized signal value of anomalous transaction amounts. Addressing these findings through targeted feature engineering, channel-level controls, and adaptive machine-learning pipelines will significantly improve detection rates while minimizing false-positive friction for legitimate customers. Continued monitoring and periodic re-analysis are essential as fraud patterns evolve.